DrChatPatin

User Guide

Medical Chatbot for Differential Diagnosis

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1. System Requirements

The following requirements must be met before accessing DrChatPatin. The application is available at:

`https://zu6ru-giaaa-aaaad-qhqba-cai.icp0.io`

The platform is accessible from any modern web browser on desktop and mobile devices. No software installation is required.

Access requirements:

- Recommended browsers: Google Chrome or Brave.
- Pop-up windows must be enabled for the application domain (required for NFID authentication).
- An active internet connection.
- Compatible with desktop and mobile devices (smartphone or tablet). No app installation required; access directly through the browser.

> **Note:** *Chrome or Brave are strongly recommended. Other browsers may not support the NF ID authentication pop-up.*

Mobile access: DrChatPatin is also available on smartphones and tablets. Open the URL in the Brave or Chrome mobile browser to access the full interface. The figure below shows the application running on a mobile device.

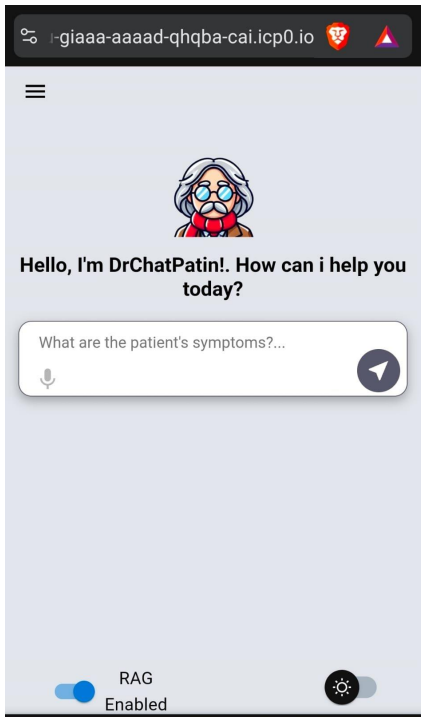

Fig.1. *DrChatPatin running on a mobile device (Brave browser).*

1.1 Enabling Pop-up Windows

On first access, the browser may block pop-up windows. Follow the steps below to allow pop-ups for the DrChatPatin domain.

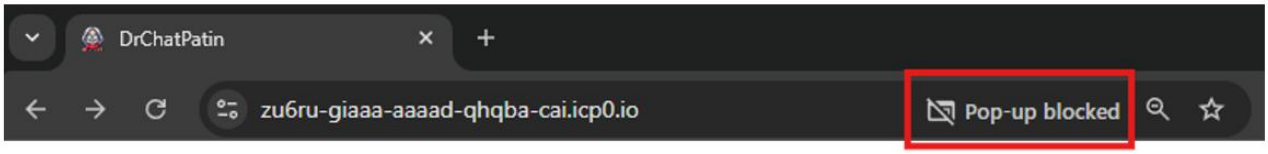

Fig 2

Step 1. Click the pop-up blocked icon in the browser address bar (shown above).

Step 2. The following dialog will appear:

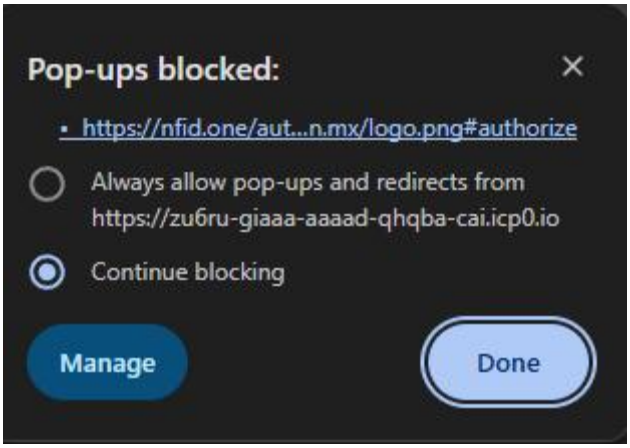

Fig. 3.

Step 3. Select "Always allow pop-ups and redirects" and click "Done".

Step 4. Reload the page. The NF ID authentication window will open automatically.

2. Authentication

DrChatPatin uses NFID, an authentication service available to ICP applications, for user authentication. Four authentication methods are available:

Authentication Method	Description
Email (Recommended)	Requires only an active email address. No password needed. A verification link is sent to the address provided.
Passkey	Requires a local password manager browser extension or an external mobile device.
Google	Uses a Google account to sign in. The email is linked to NF ID for identity generation. No personal information is accessed.
Internet Identity	The primary identity provider on the ICP blockchain. Generates a cryptographic identity from a seed phrase.

To sign in using the email method (recommended):

Step 1. After enabling pop-ups and reloading the page, the NF ID window will appear:

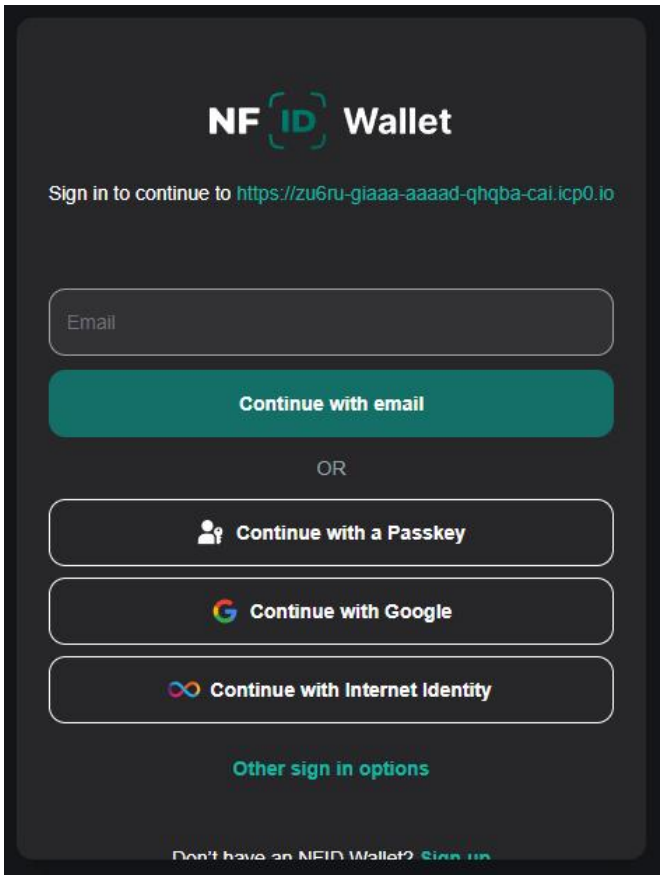

Fig. 4.

Step 2. Enter your email address and click "Continue with email".

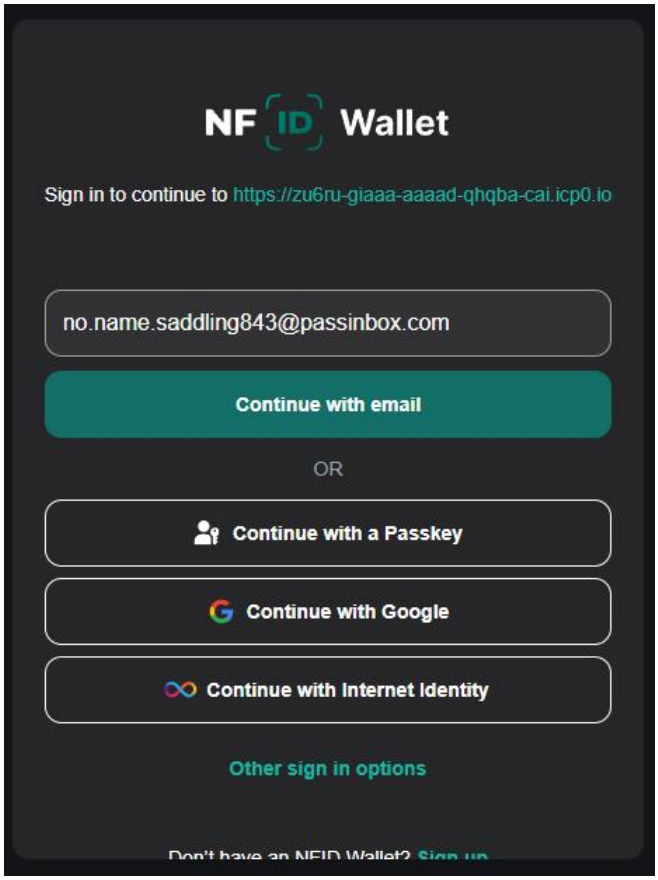

Fig. 5.

Step 3. A verification email will be sent. The following confirmation screen will appear:

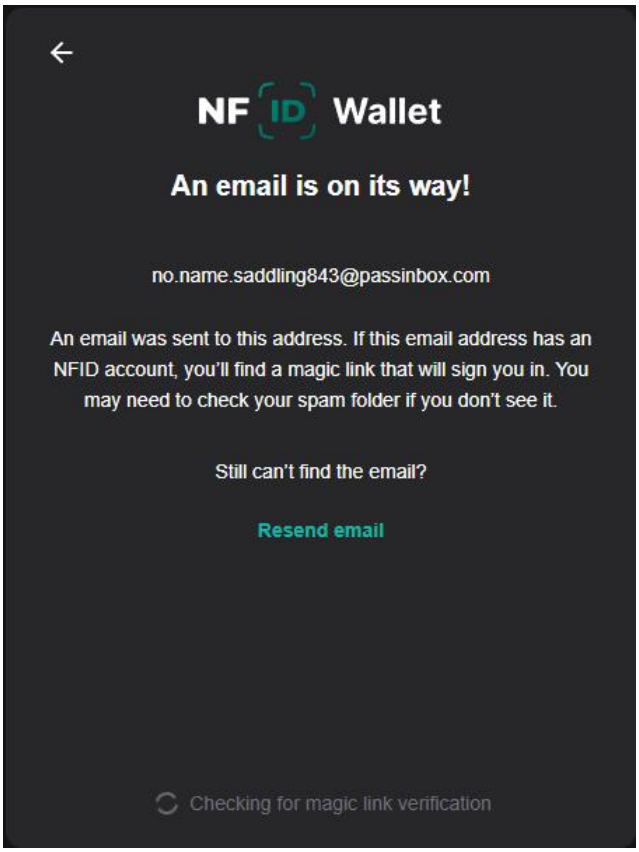

Fig 6.

Step 4. Open the verification email and click "Verify email".

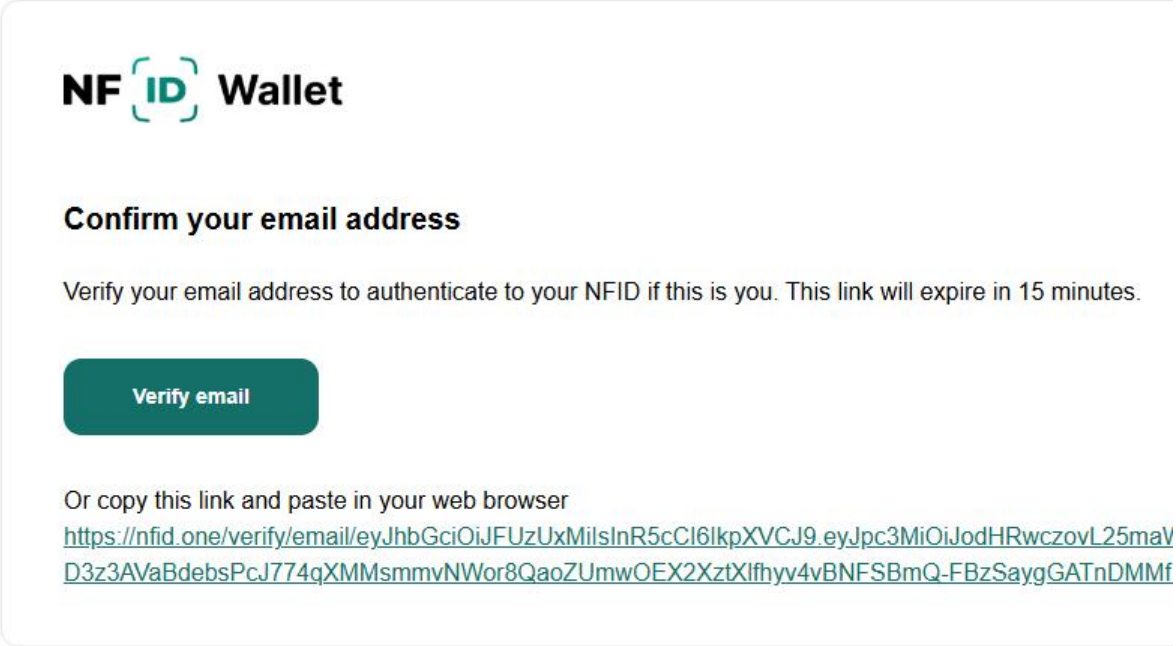

Fig.7.

Step 5. After verification, click "Continue" on the NF ID confirmation page.

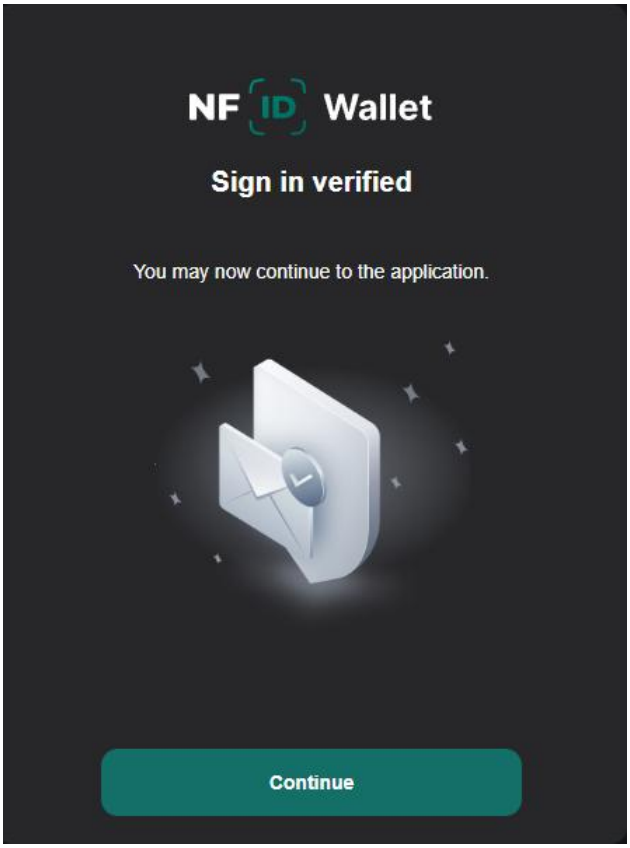

Fig. 8.

Step 6. On the connection screen, the default option "Connect anonymously" is pre-selected. Click "Connect" to proceed.

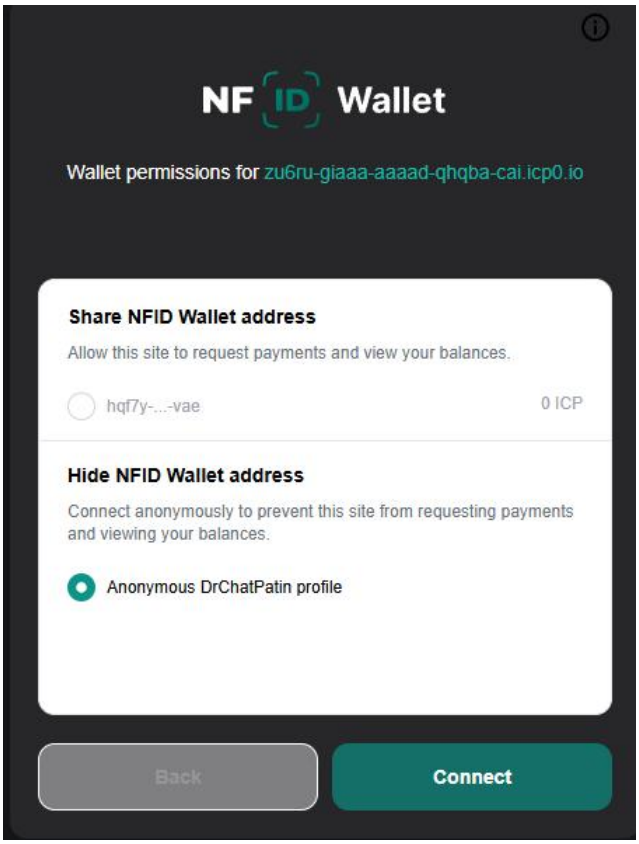

Fig. 9.

> **Note:** *The ICP identity is used solely as a user identifier and for encrypted session storage. No digital assets are required or accessed.*

3. Interface Overview

After successful authentication, the main DrChatPatin interface will load:

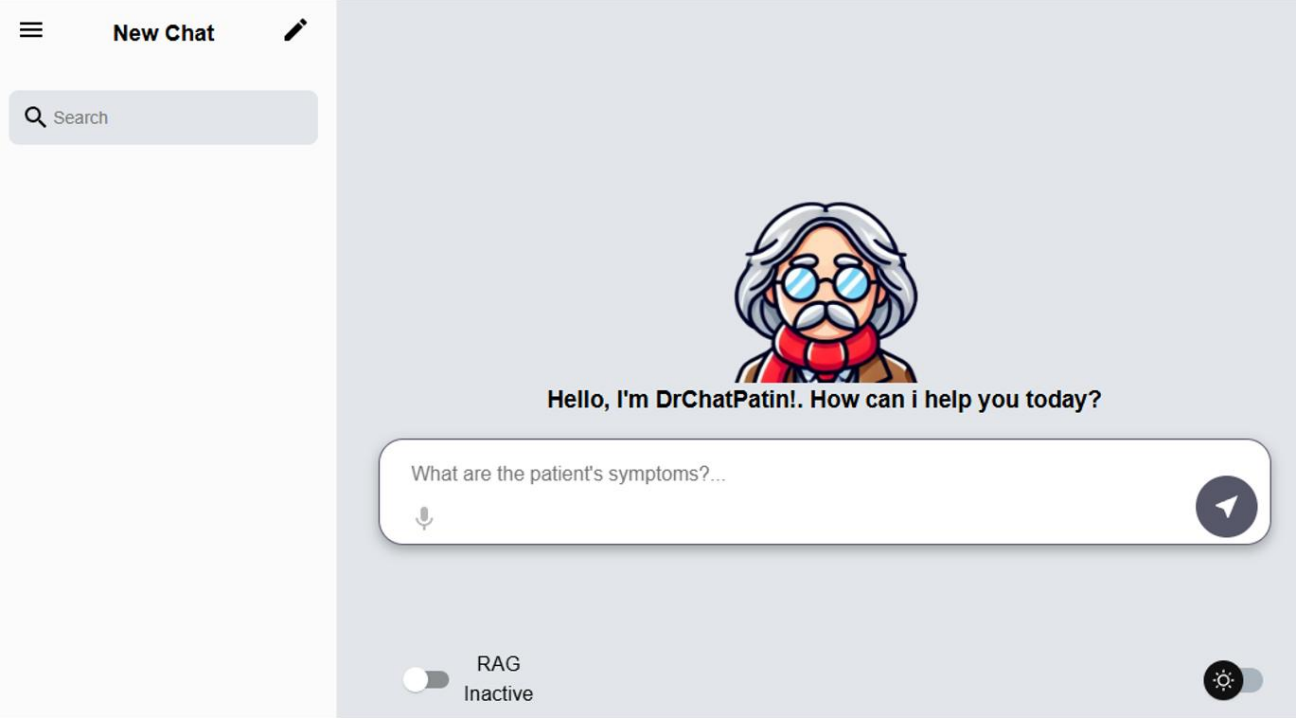

Fig. 10.

The interface provides four clearly identified controls, each highlighted by a colored border in Figure 9. The blue border marks the sidebar toggle button, which shows or hides the conversation history panel on the left. The green border identifies the New Chat button, which starts a new conversation and is only active within an existing session. The orange border indicates the Speech-to-Text input button, which activates the microphone for voice input (available in English only). The red border highlights the RAG toggle, which enables or disables retrieval from the NORD knowledge base for the current session. Additionally, the light/dark mode switch is located in the lower-right corner of the interface.

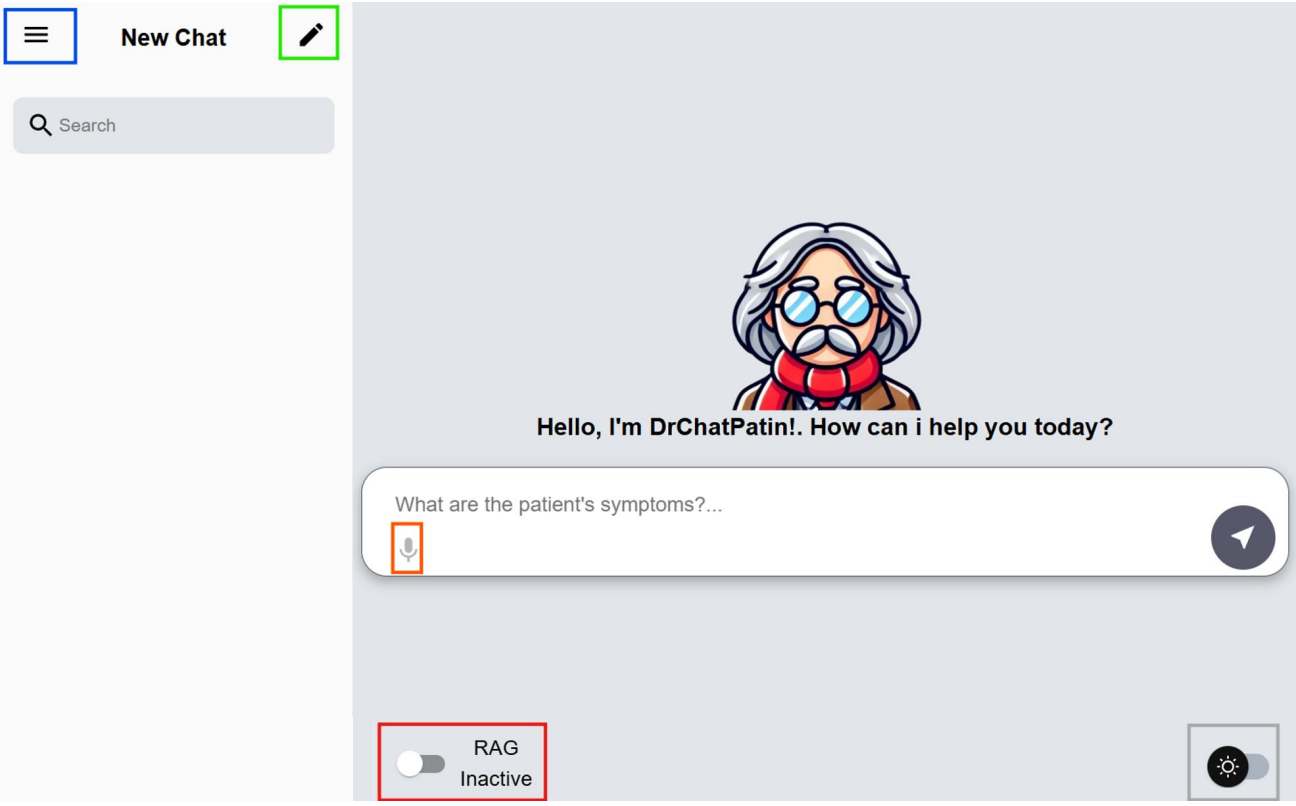

Fig. 11.

4. Conducting a Clinical Consultation

Confidentiality notice: To protect patient privacy, do not enter directly identifiable personal data into DrChatPatin. This includes full names, exact dates of birth, addresses, phone numbers, email addresses, medical record numbers, national identification numbers, or any other information that could directly identify the patient. Only the clinical information required for differential diagnosis should be provided.

4.1 Selecting the Consultation Mode

Before starting a new conversation, select the consultation mode using the **RAG toggle** in the lower-left corner:

- RAG Enabled: the system retrieves relevant fragments from the NORD rare disease knowledge base before generating the differential diagnosis. Recommended for cases with suspected rare conditions.
- RAG Disabled: the system generates the differential diagnosis directly from the model's pretraining knowledge. Generally suitable for common disease presentations.

> **Note:** *Do not switch between RAG Enabled and RAG Disabled within the same conversation. Changing the mode mid-session may produce unexpected results. Start a new conversation if a mode change is needed.*

4.2 Submitting the Clinical Case

Enter the full clinical case description in the text field at the bottom of the interface. Include all available data: symptoms, duration, relevant medical history, and any laboratory or imaging findings.

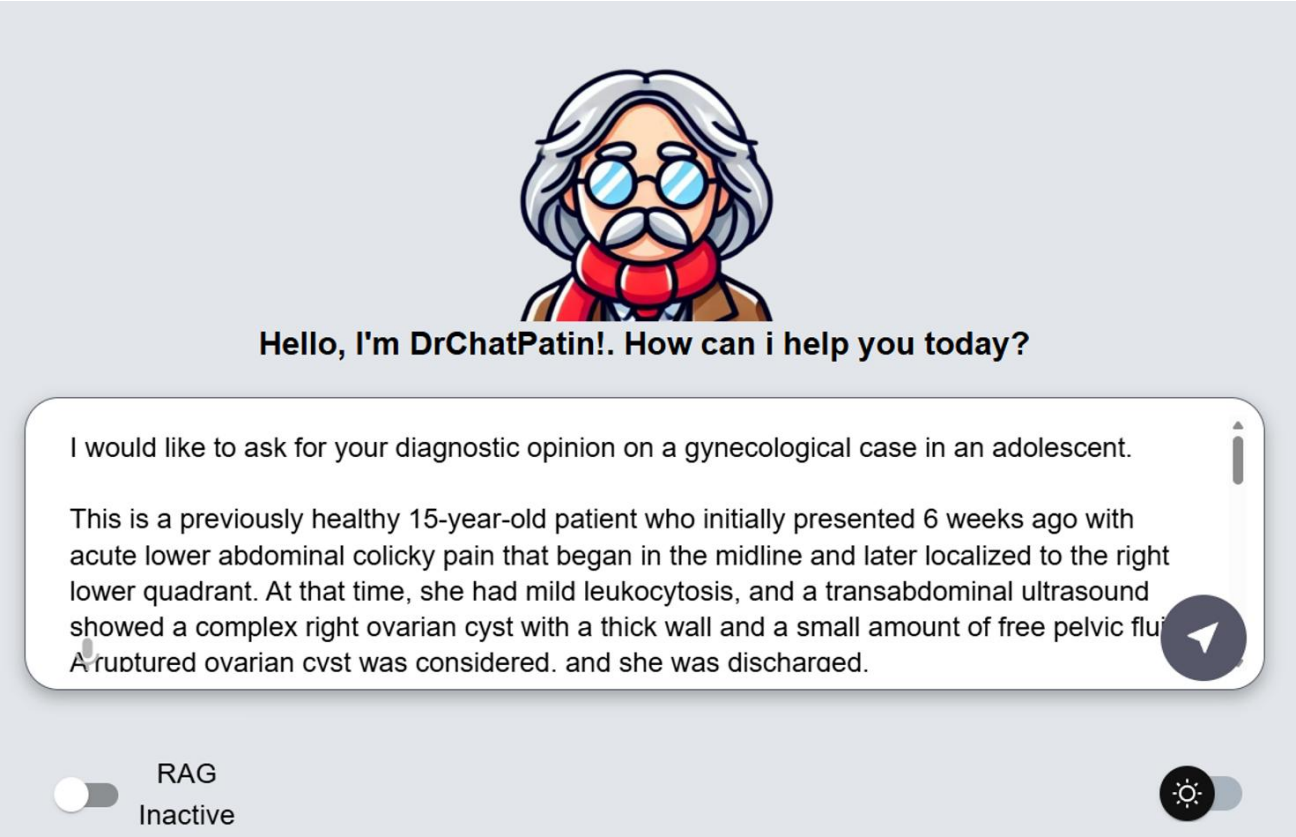

Fig. 12.

4.3 Responding to Follow-up Questions

The system will identify missing or clinically relevant information and ask focused follow-up questions. Provide the requested information if available, or indicate that no further data can be provided.

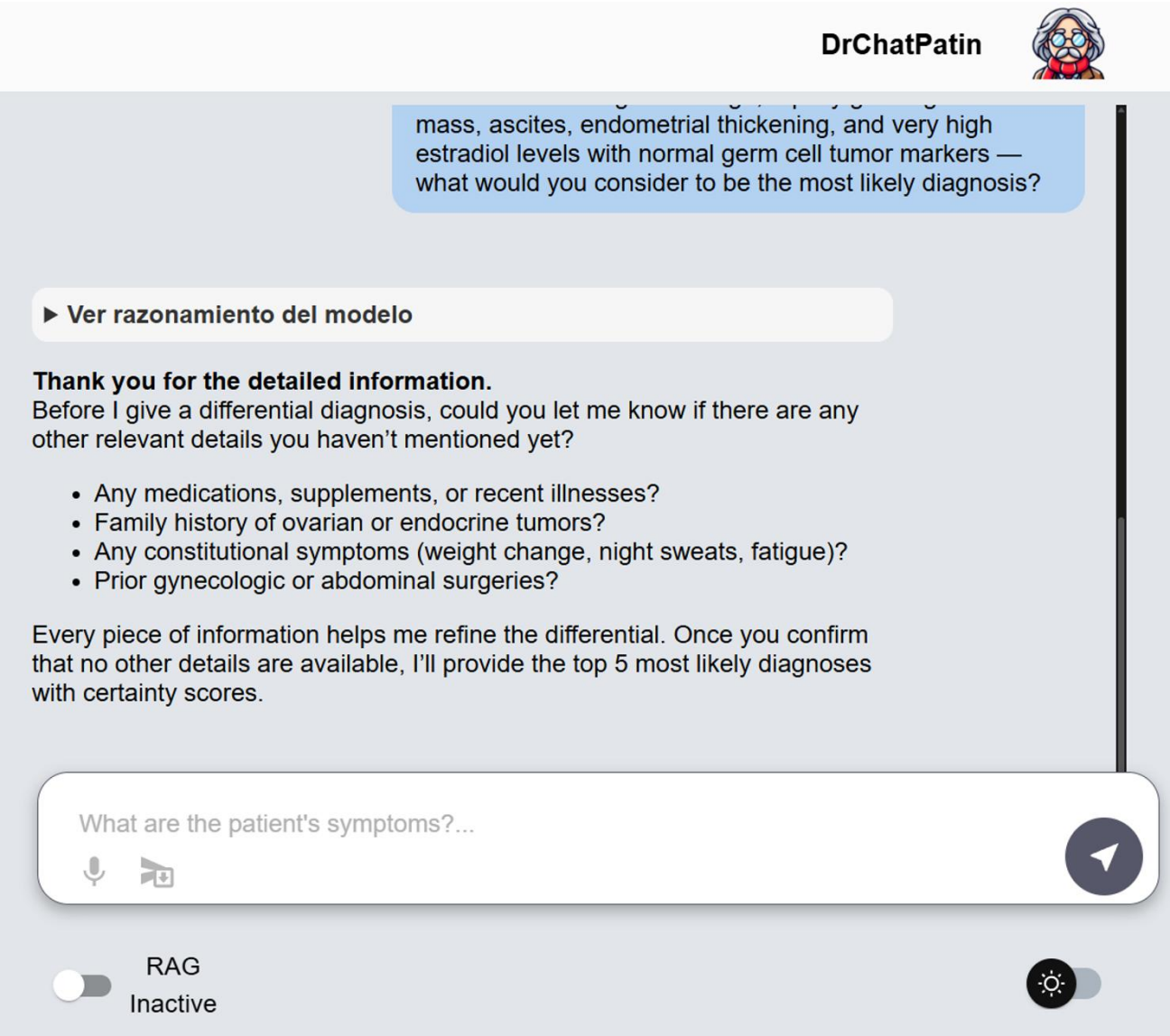

Fig. 13.

If no additional information is available, use the following standardized response:

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"There is no more information available, proceed to differential diagnosis."
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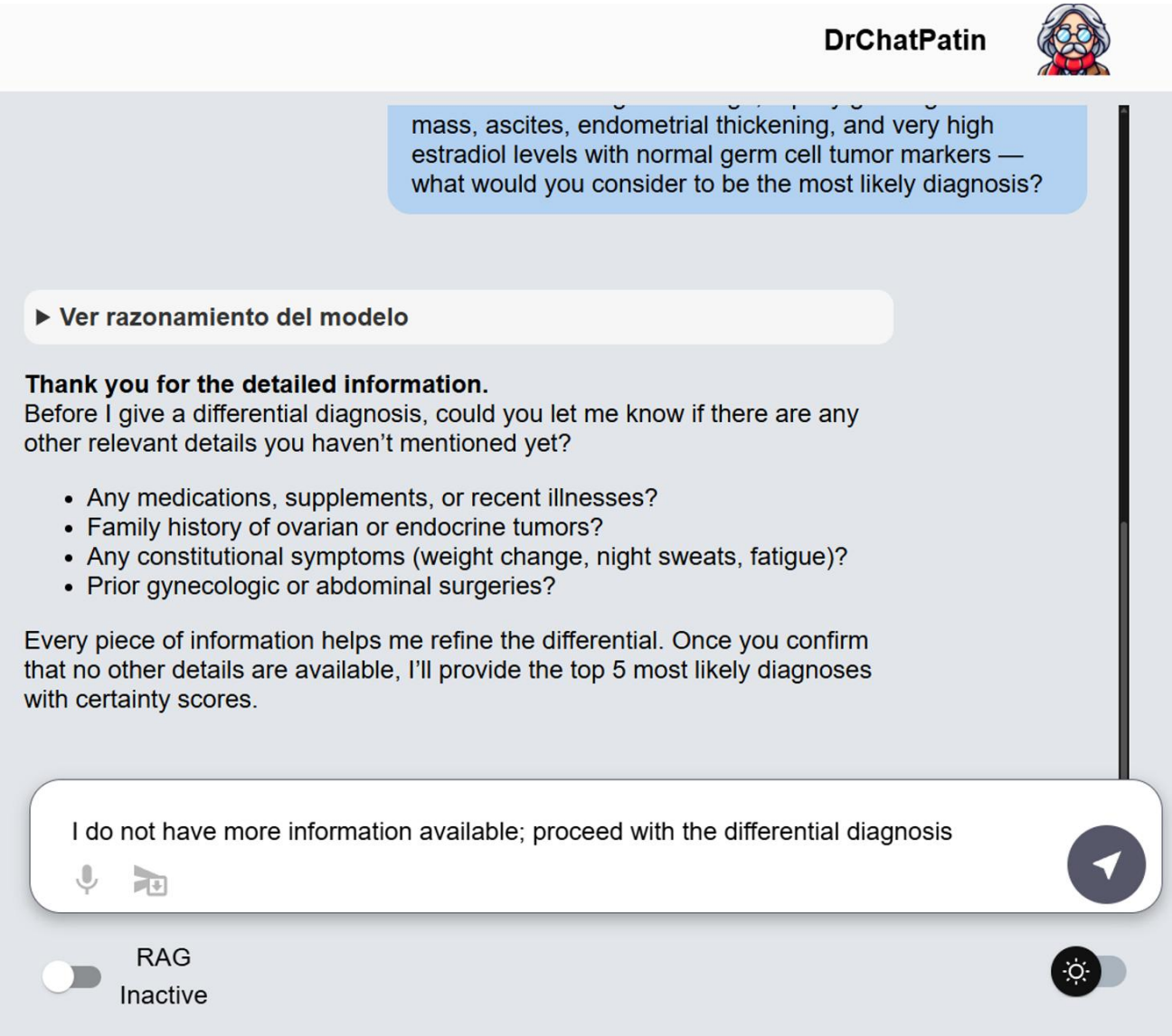

Fig. 14.

4.4 Reviewing the Differential Diagnosis

Once the system has sufficient clinical context, it will generate a ranked top-5 differential diagnosis list. Each entry includes a confidence score and a brief clinical rationale. When RAG mode is enabled, the retrieved NORD Disease Reports are listed at the bottom of the response under "Diseases cited".

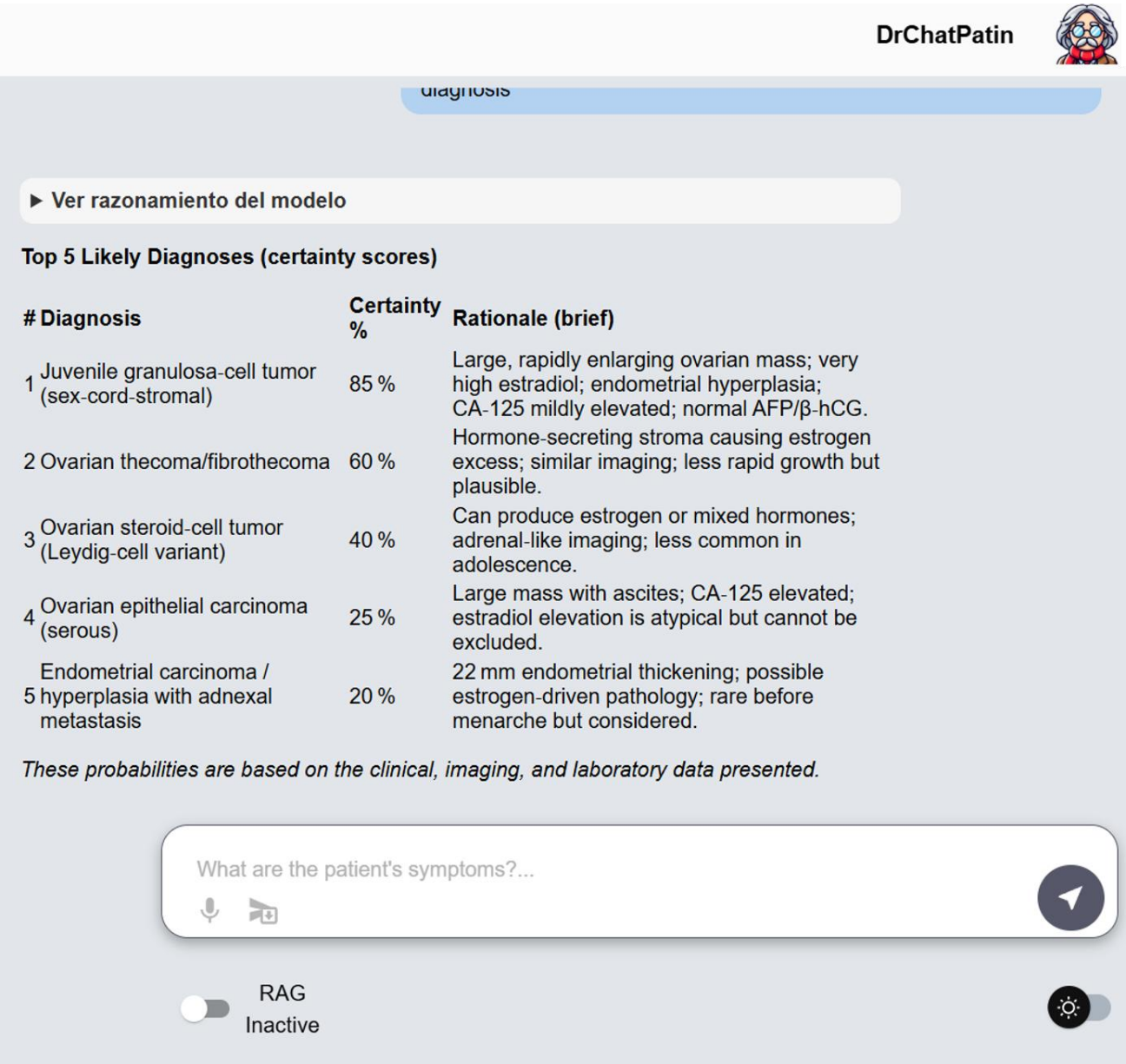

Fig.15.

5. Technical Notes

- All clinical interactions are stored as AES-256 encrypted sessions on the ICP blockchain. Clinical data is encrypted in transit, and conversation records are stored as AES-256-encrypted sessions.
- Conversation sessions persist automatically and can be resumed for longitudinal case review using the sidebar.
- The Speech-to-Text feature is currently available for English only.
- Confidence scores represent relative ranking weights, not calibrated probabilities.

- DrChatPatin is a clinical decision support tool. It does not replace clinical judgment and is not intended for autonomous diagnostic use.